

Telecom Customer Churn Analysis

End-to-end data analytics portfolio project using Python, SQL, and Tableau

- Cleaned and prepared a telecom customer dataset for analysis
- Used SQL to answer churn, revenue loss, and segmentation questions
- Built two Tableau dashboards for churn behavior and financial impact
- Translated findings into retention and win-back recommendations

50.84%

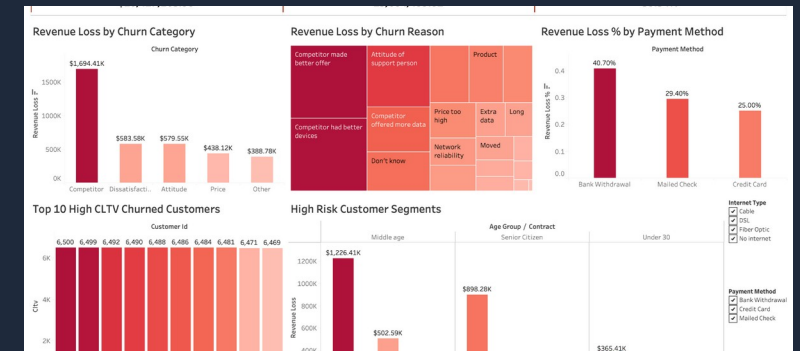
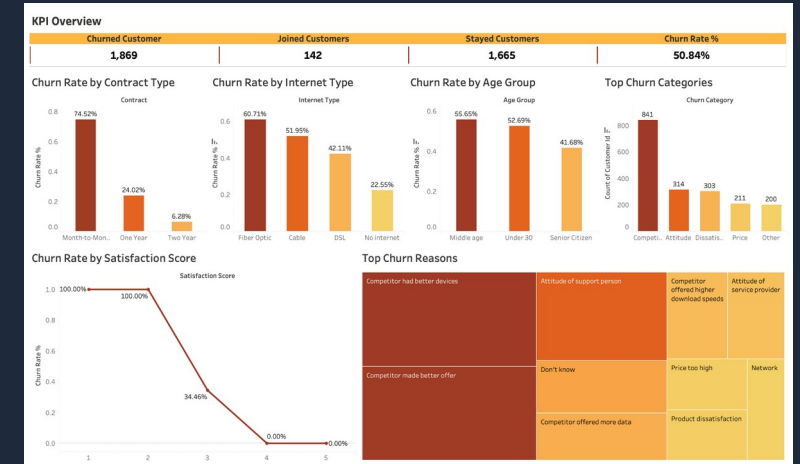
Overall churn rate

\$3.68M

Revenue lost

35.34%

Revenue loss %



Python

SQL

Tableau

Aditya Pandey

Data / Business Analyst Portfolio Project

Business problem and analytical approach

From churn symptoms to action-ready retention priorities

Business challenge

Telecom churn reduces recurring revenue and raises acquisition pressure. The analysis identifies which customer groups leave, why they leave, and where revenue loss is concentrated.

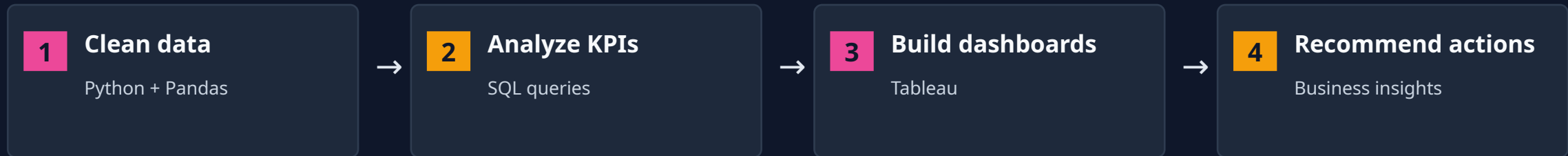
Dataset scope

3,676 customers across 44 features, including contract, internet type, satisfaction score, churn reason, total revenue, CLTV, age group, and payment method.

Outcome

A business-ready view of churn drivers, revenue impact, high-risk segments, and high-CLTV customers for targeted retention and win-back actions.

Workflow

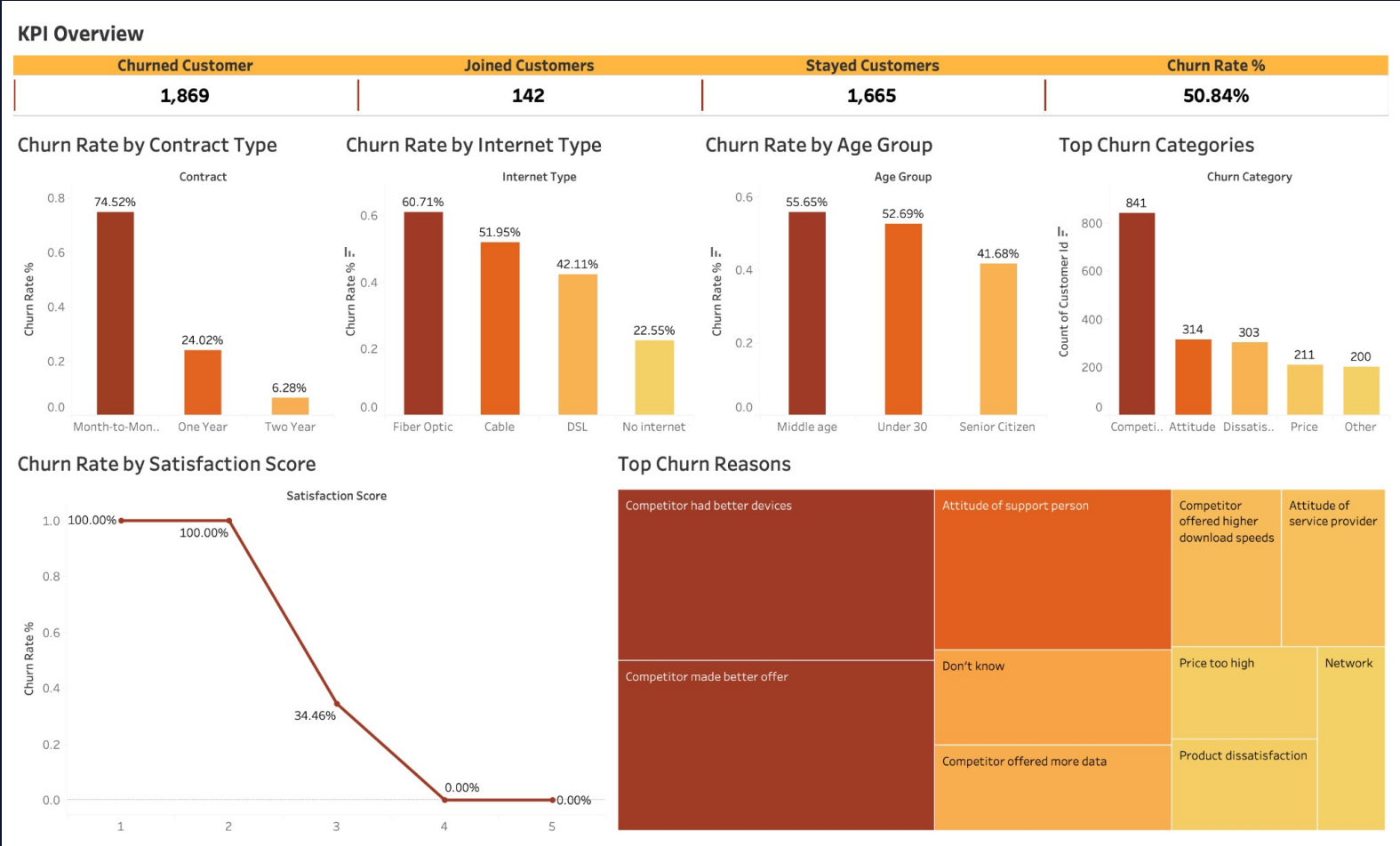


Analysis domains



Churn Analysis Dashboard

Actual Tableau dashboard screenshot with accurate KPI values



Overall churn

50.84% churn rate | 1,869 churned customers

Contract risk

Month-to-Month churn rate is 74.52%, compared with 24.02% for One Year and 6.28% for Two Year.

Internet type

Fiber Optic customers churn at 60.71%, the highest among internet types.

Customer profile

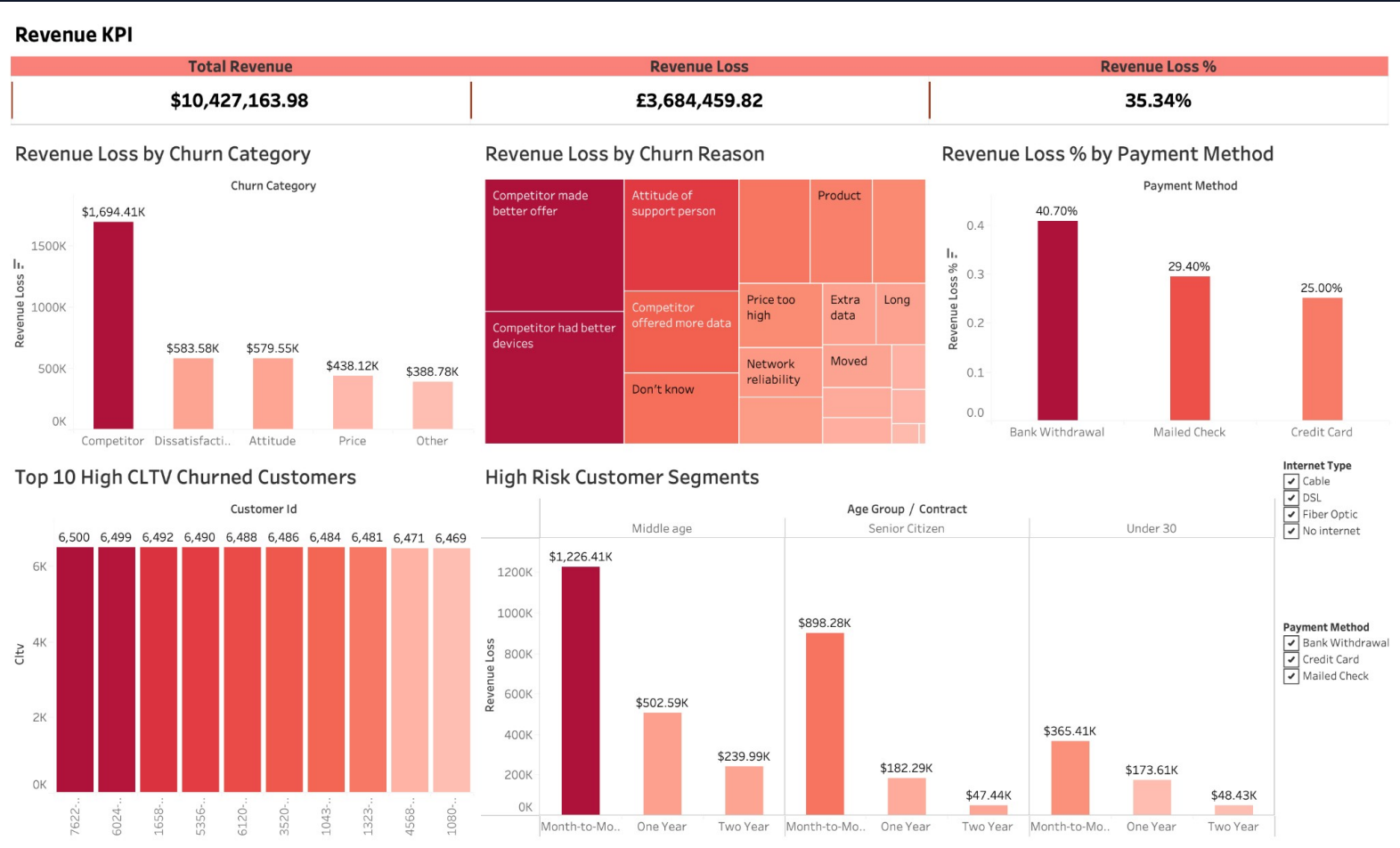
Middle-age customers show the highest churn rate at 55.65%.

Churn drivers

Competitor-related reasons and low satisfaction scores are clear warning signals.

Revenue Impact Dashboard

Actual Tableau dashboard screenshot showing financial impact of churn



Revenue at risk

Revenue lost due to churn: \$3.68M; overall revenue loss percentage: 35.34%.

Top loss category

Competitor category has the highest revenue loss, about \$1.69M.

Payment method risk

Bank Withdrawal shows the highest revenue loss percentage by payment method at 40.70%.

Win-back focus

Top high-CLTV churned customers are identified for targeted recovery offers.

High-risk segments

Month-to-Month segments drive the largest revenue-loss exposure across age groups.

Key findings

What the analysis revealed across churn behavior and revenue impact

1 Churn is high

Overall churn rate is 50.84%, with 1,869 churned customers.

2 Contracts matter

Month-to-Month contracts lead churn at 74.52%.

3 Fiber Optic risk

Fiber Optic customers churn at 60.71%.

4 Satisfaction matters

Low satisfaction scores are strongly associated with churn.

5 Competitor pressure

Competitor-related churn is the top driver and top revenue-loss category.

6 Revenue exposure

\$3.68M in revenue was lost, equal to 35.34% of total revenue.

7 Payment risk

Bank Withdrawal has the highest revenue loss percentage by payment method.

8 Win-back targets

High-CLTV churned customers should be prioritized for recovery campaigns.

Main business signal: churn is not just a volume problem; it is a revenue-protection and segment-prioritization problem.

Business recommendations

Actions linked directly to the churn and revenue evidence

Incentivize contract upgrades

Move Month-to-Month customers into One Year or Two Year plans with loyalty discounts and bundles.

Target low-satisfaction customers early

Trigger proactive support and retention offers when satisfaction scores drop.

Improve Fiber Optic experience

Review pricing, reliability, speed expectations, and support quality for Fiber Optic users.

Respond to competitor pressure

Benchmark competitor pricing, devices, and offers to reduce competitor-driven churn.

Review payment-method risk

Investigate billing friction and payment experience for higher-risk payment methods.

Prioritize high-CLTV win-back

Create personalized recovery offers for high-value churned customers.

Project structure and reproducibility

Clean repository organization for a data analyst portfolio

data/

raw_telco.csv and cleaned_telco_churn.csv for original and analysis-ready data.

notebooks/

data_cleaning.ipynb documents Python cleaning and preprocessing steps.

sql/

churn_analysis.sql answers churn KPIs, revenue loss, segmentation, and recommendations.

tableau/

telco_churn_dashboard.twb contains the Tableau workbook with both dashboards.

images/

Dashboard screenshots used in README and presentation for quick review.

README + LICENSE

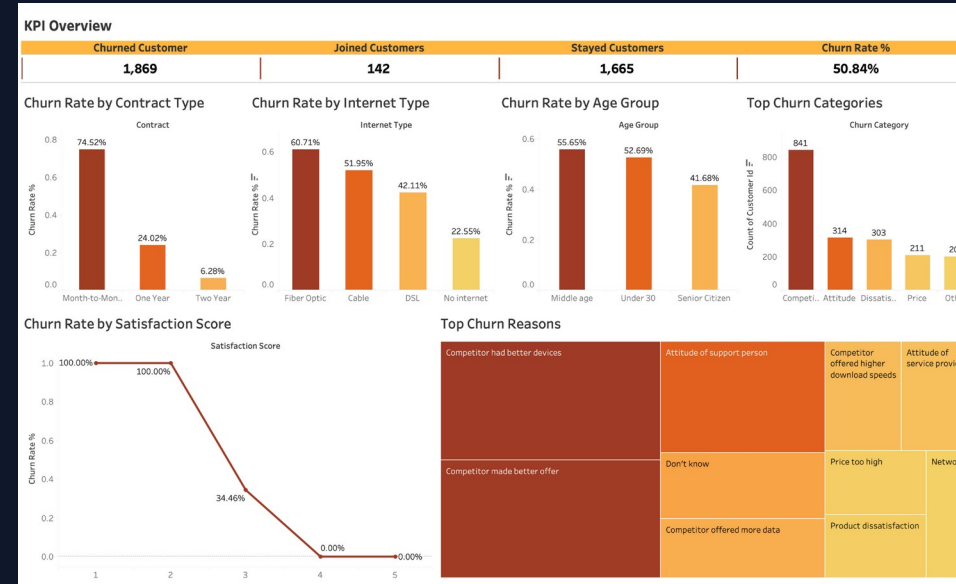
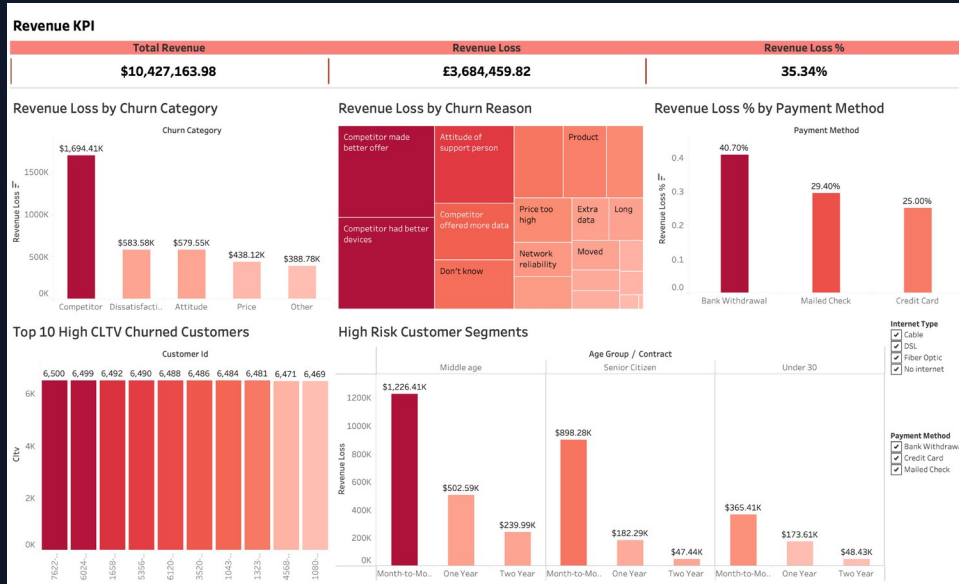
Professional documentation, setup instructions, and MIT license.

Reproducibility steps

- Install dependencies: `pip install -r requirements.txt`
- Run Python cleaning notebook in Jupyter
- Import cleaned CSV into SQL and run `churn_analysis.sql`
- Open Tableau workbook and reconnect the cleaned dataset if needed

Portfolio outcome

What This Project Demonstrates



Analytical skills

Python cleaning, SQL KPI logic, segmentation, revenue-loss analysis, and interpretation.

Business thinking

Clear recommendations tied to contract risk, satisfaction, competitor pressure, and revenue exposure.

Portfolio readiness

Clean repository, README, visuals, workbook, SQL script, data, and reproducible setup.

Prepared by Aditya Pandey

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